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CHILD RESTRAINT

Abstract

A child restraint includes a seat bottom and a seat back coupled to the seat bottom. The seat back includes a backrest configured to extend upwardly from the seat bottom and a headrest coupled to the backrest for supporting a head of a child. The headrest includes a rear section arranged along the backrest, a first side section extending outwardly away from the rear section, and a second side section extending outwardly away from the rear section.

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Background/Summary

PRIORITY CLAIM [0001] This application is a continuation of U.S. patent application Ser. No. 18/739,653, filed Jun. 11, 2024, which is a continuation of U.S. patent application Ser. No. 17/848,732, filed Jun. 24, 2022, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Application Ser. No. 63/215,069, filed Jun. 25, 2021, each of which is expressly incorporated by reference herein.

BACKGROUND

[0002] The present disclosure relates to a child restraint, and particularly to a child restraint configured to be secured to a vehicle seat within a vehicle. More particularly, the present disclosure relates to a child restraint that is configured to reduce undesirable forces acting on a child in the child restraint during operation of vehicle.

SUMMARY

[0003] According to the present disclosure, a child restraint includes a seat bottom and a seat back coupled to the seat bottom. The seat back includes a backrest configured to extend upwardly from the seat bottom and a headrest coupled to the backrest for supporting a head of a child. The headrest includes a rear section arranged along a forward-facing surface of the backrest, a first side section extending outwardly away from the rear section, and a second side section extending outwardly away from the rear section and away from the first side section.

[0004] In illustrative embodiments, the child restraint further includes energy-redirection means coupled to the first and second side sections. The energy-redirection means redirect at least a portion of a cumulative force from the head of the child during an impact event into at least a first force extending into one of the first and second side sections and a second force extending outwardly away from the rear section of the headrest so that the head of the child does not experience the cumulative force when impacting the one of the first and second side sections. The first and second forces are each less than the cumulative force as a result of the head of child impacting the energy-redirection means.

[0005] In illustrative embodiments, the energy-redirection means includes at least one energy-redirection layer that is coupled to both the first side section and the second side section. The energy redirection layer has an outer, energy-redirection surface that faces toward the head of the child. The energy-redirection surface may be convexly-shaped relative to the head of the child. During a lateral impact event, the energy-redirection layer causes the head of the child to roll across the energy-redirection surface and to maintain contact with the energy-redirection surface for a longer period of time and/or distance during the lateral impact event. This disperses forces acting on the head of the child over a greater area and longer period of time thereby reducing the potential for a cumulative force that may injure the child.

[0006] Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of illustrative embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0007] The detailed description particularly refers to the accompanying figures in which:

[0008] FIG. **1** is a perspective view of a child restraint, in accordance with the present disclosure, including a seat bottom and a seat back coupled to the seat bottom, the seat back including a backrest and a headrest, and showing that the headrest includes a rear section and left and right side sections, each side section including a support base, a comfort layer, and an energy-redirection layer or insert that may be added to the headrest to increase side impact force dampening by providing energy-redirection means for a head of a child during a collision event;

[0009] FIG. **2** is a perspective view of the headrest shown in FIG. **1** without the energy-redirection layer and suggesting that a side impact experienced by the child may result in a cumulative lateral force on the side wing of the headrest;

[0010] FIG. **3** is a perspective view of the headrest shown in FIG. **1** with the energy-redirection layer installed on the headrest and suggesting that the energy-redirection layer is configured to redirect or separate forces imparted on the side wing by the child's head into at least a first force extending laterally into the side section of the headrest and a second force extending outwardly away from the rear section of the headrest so that the child does not experience the cumulative force when impacting the side section;

[0011] FIG. **4** is a perspective view of one of the energy-redirection layers showing that the energy-redirection layer includes a convex outer surface configured to support the head of the child during the lateral collision event, a rear surface configured to interface with one of the side sections of the headrest, and a chamfered surface configured to interface with the rear section of the headrest;

[0012] FIG. **5** is a rear view of the energy-redirection layer of FIG. **4**;

[0013] FIG. **6** is a cross-sectional view of the energy-redirection layer taken along line **6-6** in FIG. **4** showing that the energy-redirection layer has a thickness that decreases from an upper end of the energy-redirection layer to a lower end of the energy-redirection layer;

[0014] FIG. **7** is a cross-sectional view of the energy-redirection layer taken along line **7-7** in FIG. **4** showing the shape of the energy-redirection layer that provides the convex outer surface, the rear surface, and the chamfered surface;

[0015] FIG. **8** is a cross-sectional view of the headrest from FIGS. **1-3** showing the support layer, the comfort layer, and the energy-redirection inserts interfacing with one another;

[0016] FIG. **9** is a cross-sectional view of another headrest, in accordance with the present disclosure, including a support layer and a one-piece, energy-redirection insert that provides both comfort and energy-redirection means for the head of a child; and

[0017] FIG. **10** is a cross-section view of another headrest, in accordance with the present disclosure, which is formed as one piece from a plastic material and that is formed to include a left and a right side section having an outwardly-facing convex surface that provides energy-redirection means for the head of a child

[0018] FIG. **11** is a perspective view of another child restraint, in accordance with the present disclosure, the child restraint including a seat bottom and a seat back having a backrest and an integrated headrest that is fixed to the backrest and includes left and right energy-redirection units for redirecting the child's head during a lateral collision event to minimize forces acting on the child's head from the child restraint; and

[0019] FIG. **12** is another perspective view of the child restraint from FIG. **11**.

DETAILED DESCRIPTION

[0020] A first embodiment of a child restraint **10** in accordance with the present disclosure is shown in FIGS. **1-3**. The child restraint **10** includes a headrest **20** including energy-redirection means or inserts **34** as shown in FIGS. **1-8**. A second embodiment of a headrest **220** in accordance with the present disclosure is shown in FIG. **9**. A third embodiment of a headrest **320** in accordance with the present disclosure is shown in FIG. **10**. A fourth embodiment of a headrest **420** in accordance with the present disclosure is shown in FIGS. **11** and **12**.

[0021] The child restraint **10** includes a seat bottom **12** and a seat back **14** coupled to the seat bottom **12**. The seat bottom **12** and the seat back **14** define a child-receiving space **16** to hold a child for transportation in a vehicle, for example. The seat back **14** includes a backrest **18** coupled to the seat bottom **12** and arranged to extend upwardly from the seat bottom **12** and the headrest **20** coupled to the backrest for supporting a head of the child.

[0022] The headrest **20** may include a rear section **22** coupled to the backrest **18**, a first side section or wing **24** extending outwardly away from the rear section **22**, and a second side section or wing **26** extending outwardly away from the rear section **22** and away from the first side section **24**. In some embodiments, the headrest **20** is movable relative to the backrest **18**. In some embodiments, the headrest **20** forms an upper part of the backrest **18** and is integral therewith (i.e., fixed in position relative to the backrest).

[0023] Each side section **24**, **26** includes a support layer **30**, a comfort layer **32**, and an energy-redirection layer or insert **34** as shown in FIG. **4**. The support layer **30** is illustratively a polymeric panel arranged between the backrest **18** and the comfort layer **32**. The comfort layer **32** is, for example, a foam layer that is configured to provide cushioning for the head of a child. The support layer **30** and the comfort layer **32** establish an outer contour **40** of each side section **24**, **26** that may interact with the head of a child during a collision event to absorb energy from the head of a child when the headrest does not include the energy-redirection layer **34**. The energy-redirection layer **34** is configured to replace or rest on top of the comfort layer **32** and establish a second outer contour **42** of the first and second side sections **24**, **26** different than the outer contour **40** provided by the comfort layer **32**.

[0024] The energy-redirection layer **34** is configured to provide energy-redirection means for redirecting at least a portion of a cumulative force **50** from the head of the child during an impact event, for example a lateral collision event, into at least a first force **52** extending into one of the first and second side sections **24**, **26** and a second force **54** extending outwardly away from the rear section **22** of the headrest **20** so that the child does not experience the cumulative force **50** when impacting one of the first and second side sections **24**, **26**. The cumulative force **50** extends in the same direction as first force **52** and has a higher magnitude than first force **52** and second force **54**. In some embodiments, cumulative force **50** is split or transformed into first force **52** and second force **54**. First force **52** and second force **54** may satisfy safety standards while cumulative force **50** would not have satisfied the same safety standards.

[0025] The energy-redirection layer **34** may be an insert that is fitted between the comfort layer **32** and an outer trim **38** of the headrest **20** or between the support layer **30** and the outer trim **38**. The child restraint **10** may be retrofitted with the energy-redirection layer **34**. Each energy-redirection layer or insert **34** is formed from a material (i.e. a plastic or foam, such as, expanded polyethylene (EPE), expanded polypropylene (EPP), porous expanded polypropylene (P-EPP), cross-linked expanded polyethylene (xEPE), etc.) that does not collapse or compress substantially during a collision event so that the head of the child rolls along an outer surface **60** of the energy-redirection layer **34** during the collision event. In some embodiments, the headrest **20** also does not include any side impact air bags which tend to compress with load and may not direct the head of the child to roll along the outer surface **60**.

[0026] Each of the energy-redirection layers **34** may have the following properties and/or characteristics shown in Table 1.

TABLE-US-00001 TABLE 1 Physical P- Property Unit EPE EPP xEPE EPP Average pcd 1 1.3 1.5 1.9 2.8 1 1.3 1.9 2.8 1.5 2.8 Density Compression psi 8 10 11 13 22 11 14.5 23.5 42 6 23 Strength @ 25% Compression psi 16 18 19 22 35 19 23.5 33.5 54 15 35 Strength @ 50% Compression psi 38 44 49 56 75 41 45 64 111 40 79 Strength @ 75% Tensile % 39 40 45 52 70 35 38 55.5 67 22 27 Strength Tensile lbs/in 38 32 30 29 25 18 16 15 14 50 13 Elongation Tear Strength % 12 14 16 17 21 9 10 13 16 12 19 Comp Set @ % 3 3 4 4 4 8 8 7 7 2 5 25% Comp Set @ % 12 14 13 12 12 16 14 12 12 6 9 50% Buoyancy pcf 61.2 60.6 59.5 59.5 59.1 61 60.5 59.5 59 61 n/a Thermal (K) 0.26 0.26 0.24 0.24 0.24 0.24 0.24 0.24 0.24 0.25 0.265 Conductivity BTU- in/ ft-hr-f Thermal (R) @ 3.9 4 4.2 4.2 4.2 4.2 4.2 4.2 4.2 4 3.8 Resistance 70 F. Service Temp F. 160 160 160 160 160 212 212 212 212 185 212 Water % −1 −1 −1 −1 −1 −1 −1 −1 −1 −1 −1 Absorption Compression 1000 hr 2.5 3 3 3.3 3 n/a n/a n/a n/a n/a n/a Creep @ 1 psi

[0027] The properties and/or characteristics in Table 1 above were calculated using the test methods shown in Table 2 at the time of filing this patent application.
TABLE-US-00002 TABLE 2 Physical Property Test Method Average Density ASTM-D3575 Comp Strength @ 25% ASTM-D3575 Comp Strength @ 50% ASTM-D3575 Comp Strength @ 75% ASTM-D3575 Tensile Strength ASTM-D3575 Tensile Elongation ASTM-D3575 Tear Strength ASTM-D3575 Comp Set @ 25% ASTM-D3575 Comp Set @ 50% ASTM-D3575 Buoyancy ASTM-D3575 Thermal Conductivity ASTM-C177 Thermal Resistance ASTM-C178 Service Temp ASTM-D3575 Water Absorption ASTM-D3575 Comp Creep ASTM-D3575
[0028] Each of the materials shown in Table 1 also passed a Flammability test according to test method FMVSS-302. Each of the materials shown in Table 1 also passed a Fuel Immersion test according to test method Coast Guard (CGD-770145) Fuel B.
[0029] Each energy-redirection insert **34** may include a convex, outer, energy-redirection surface **60** providing the second outer contour **42**, a rear surface **62** matching the outer contour **40**, and a chamfered surface **64** as shown in FIGS. **6** and **7**. The outer surface **60**, for example, is curved and may cause the head of the child to roll along the outer surface **60** during the collision event to provide the first and second forces **52**, **54** rather than the cumulative force **50**. The rear surface **62** abuts against matches the outer contour **40** of the respective side section without the energy-redirection layer **34** to minimize movement of the energy-redirection layer **34** relative to the rest of the headrest **20**. The chamfered surface **64** abuts against and interfaces with the rear section **22**. In one example, the outer surface **60** may be convex relative to the head of the child and concave relative to the support layer **30**.
[0030] Each energy-redirection layer **34** or insert has a thickness **70** that changes to provide the energy-redirection means as shown in FIGS. **6** and **7**. For example, the thickness **70** of the energy-redirection layer **34** may decrease from an upper end **72** to a lower end **74** of the energy-redirection layer **34** as shown in FIG. **6**. The thickness **70** may also decrease from a rear end **76** to a forward end **78** of the energy-redirection layer **34** as shown in FIG. **7**. In some embodiments, the thickness **70** decreases constantly from the upper end **72** to the lower end **74** and/or from the rear end **76** to the forward end **78**. In some embodiments, the thickness **70** decreases gradually from the upper end **72** to the lower end **74** and/or from the rear end **76** to the forward end **78**. In some embodiments, the thickness **70** changes gradually (i.e. increases and decreases) from the upper end **72** to the lower end **74** and/or from the rear end **76** to the forward end **78**. This thickness **70** causes the head of a child to roll across outer surface **60** and to tilt downwardly during a collision event to minimize forces acting between the headrest **20** and the head of the child.
[0031] The energy-redirection layer **34** may change an angle of the outermost surface of the side sections **24**, **26** relative to rear section **22** compared to comfort layer **32** or support layer **30**. In some embodiments, an angle **65** between the outer surface **60** of each energy-redirection insert **34** and the rear section **22** is less than an angle **66** between an outer surface **33** of the comfort layer **32** and/or an angle **68** between outer surface **31** of the support layer **30** and the rear section **22** as shown in FIG. **8**. The outer surface **60** of each energy-redirection insert **34** may be substantially planar or may have a convexly-shaped outer contour **42** as suggested in FIG. **8**.
[0032] Another embodiment of a headrest **220** that can be included in child restraint **10** in place of headrest **20** and energy-redirection inserts **34** is shown in FIG. **9**. The headrest **220** is similar to headrest **20**. Accordingly, similar reference numbers in the **200** series are used to indicate similar features between headrest **220** and headrest **20**. The disclosure for headrest **20** is hereby incorporated by reference herein for headrest **220**.
[0033] The headrest **220** includes a support layer **230** and an energy-redirection layer **234** as shown in FIG. **9**. The support layer **230** may be made from a rigid plastic material, such as polypropylene sheeting or molded high impact rated resin, to support the head of a child seated on the child restraint **10**. The energy-redirection layer **234** may be made from a foam material to provide both comfort and energy-redirection means for a child's head. Both the support layer **230** and the

energy-redirection layer **234** provide portions of a rear section **222** and side sections **224**, **226** of the headrest **220**.

[0034] The energy-redirection layer **234** may replace a comfort layer(s) (i.e. comfort layer **32**) previously included in the headrest **20** while also providing energy-redirection means. The energy-redirection layer **234** has an outer, energy-redirection surface **260** that interacts with the head of the child to provide the comfort and the energy-redirection means. The outer surface **260** of the energy-redirection layer **234** is convexly-curved to gradually decrease an angle of the outer surface **260** of each side section **224**, **226** relative to the rear section **222** as the outer surface **260** extends away from the rear section **222**.

[0035] A thickness **270** of the energy-redirection layer **234** decreases in each side section **224**, **226** as the energy-redirection layer **234** extends away from the rear section **222**. The thickness **270** may decrease gradually and/or constantly from the rear section **222** to a distal end of each side section **224**, **226**. The thickness **270** may change gradually from the rear section **222** to the distal end of each side section **224**, **226**. For example, the thickness **270** may first increase from the rear section **222** to a point about midway between the rear section **222** and the distal end and then decrease from the point to the distal end of each side section **224**, **226**. In some embodiments, the outer surface **260** of the energy-redirection layer **234** is substantially planar.

[0036] Another embodiment of a headrest **320** that may be included in child restraint **10** in place of headrest **20** and energy-redirection inserts **34** is shown in FIG. **10**. The headrest **320** is similar to headrest **20**. Accordingly, similar reference numbers in the **300** series are used to indicate similar features between headrest **320** and headrest **20**. The disclosure for headrest **20** is hereby incorporated by reference herein for headrest **320**.

[0037] The headrest **320** is formed as a one-piece component that includes a rear section **322** and left and right side sections **324**, **326** as shown in FIG. **10**. The headrest **320** may be made from a rigid plastic material to support the head of a child seated on the child restraint **10** and that is shaped to provide energy-redirection means for the head of the child. In some embodiments, the headrest **320** may be made from a foam material to provide both comfort and energy-redirection means for a child's head or combinations of foam and rigid plastic material.

[0038] The side sections **324**, **326** each have an outer, energy-redirection surface **360** that interacts with the head of the child to provide the comfort and/or the energy-redirection means. The outer surface **260** of the headrest **320** is convexly-curved to gradually decrease an angle of the outer surface **360** of each side section **324**, **326** relative to the rear section **322** as the outer surface **360** extends away from the rear section **322**.

[0039] A thickness **370** of each side section **324**, **326** decreases as each side section **324**, **326** extends away from the rear section **322**. The thickness **270** may decrease gradually and/or constantly from the rear section **322** to a distal end of each side section **324**, **326**. The thickness **370** may change gradually from the rear section **322** to the distal end of each side section **324**, **326**. For example, the thickness **370** may first increase from the rear section **322** to a point about midway between the rear section **322** and the distal end and then decrease from the point to the distal end of each side section **324**, **326**. In some embodiments, the outer surface **360** of each side section **324**, **326** is substantially planar.

[0040] Another embodiment of a child restraint **400** with an integrated headrest **420** is shown in FIGS. **11** and **12**. Similar reference numbers in the **400** series are used to indicate similar features between child restraint **400** and child restraint **10**. The disclosure for child restraint **10** is hereby incorporated by reference herein for child restraint **400**.

[0041] The child restraint **400** includes a seat bottom **412** and a seat back **414**. The seat back **414** includes a backrest **418** and a headrest **420** coupled to the backrest **418**. The headrest **420** is integrated into backrest **418** of the child restraint **400** so as to be mounted to the backrest **418** in a fixed position. The headrest **420** includes first and second side sections **424**, **426**. Each side section **424**, **426** has a convexly-shaped, energy-redirection surface **460** that interacts with the head of the child. The energy-redirection surface **460** provides an outer contour **442** of the headrest that minimizes forces acting on the head of the child during a collision event. The headrest **420** may be covered with a trim and/or soft goods or foam to increase comfort for the child.

[0042] The child restraints **10**, **200** were tested in a side impact simulator to determine head injury criterion of the child. The child restraints having side section outer contour **40** received a Head Injury Criterion (HIC) score of about 700. Unexpectedly, changing the outer contour **40** to second outer contour **42** using energy-redirecting layer surface **60**, **260**, **360**, **460** decreased the HIC score by more than 50% to about 321. It was previously thought that providing a surface such as surface **60**, **260**, **360**, **460** would adversely affect the HIC score.

Claims

1. A child restraint comprising a backrest and a headrest coupled to the backrest for supporting a head of a child, the headrest including a rear section arranged along the backrest, a first side section extending outwardly away from the rear section, and a second side section extending outwardly away from the rear section, the first and second side sections spaced apart from a longitudinal centerline of the headrest located between the first and second side sections and extending from the rear section away from the backrest, wherein the first side section and the second side section each include an outer energy-redirection surface spaced a distance from the longitudinal centerline of the headrest, and wherein the distance increases as the energy-redirection surface extends away from the rear section to a distal end of each corresponding energy-redirection surface, and wherein each energy-redirection surface is included in an energy-redirection layer configured to disperse forces acting on the head of the child during a side impact event, and each energy-redirection layer has a compression strength at 25% of at least 6 psi when measured in accordance with ASTM-D3575.

2. The child restraint of claim 1, wherein each side section further includes a support layer underlying the energy-redirection surface, and the support layer of the first and second side sections have a first outer contour and the energy-redirection surface of the first and second side sections have a second outer contour, different than the first outer contour.

3. The child restraint of claim 2, wherein the second outer contour includes at least a portion that is convex relative to the head of the child.

4. The child restraint of claim 3, wherein the second outer contour is convex relative to the head of the child from the rear section to the distal end of the energy-redirection surface.

5. The child restraint of claim 2, wherein the support layer has an outer surface configured to face the head of the child, and wherein, at a location along each side section, the outer surface of the support layer is arranged at a first angle relative to a plane extending along the rear section and the energy-redirection surface is arranged at a second angle relative to the plane, the second angle being greater than the first angle.

6. The child restraint of claim 5, wherein the second angle gradually decreases along at least a portion of the energy-redirection surface as each side section extends from the rear section to the distal end.

7. The child restraint of claim 1, wherein each energy-redirection layer has an average density of at least 1 pcf.

8. The child restraint of claim 1, wherein each energy-redirection layer has an upper end and a lower end, and wherein the distance increases from the upper end to the lower end.

9. The child restraint of claim 1, wherein the headrest does not include any side impact air bags.

10. The child restraint of claim 1, wherein the headrest is movable relative to the backrest.

11. A child restraint comprising a backrest and a headrest coupled to the backrest for supporting a head of a child, the headrest including a rear section arranged along the backrest, a first side section extending outwardly away from the rear section, and a second side section extending outwardly away from the rear section, wherein the first side section and the second side section each include a support layer coupled to the backrest, a comfort layer coupled to an outer surface of the support layer, an insert configured to at least partially overly the comfort layer to position at least a portion of the comfort layer between support layer and at least a portion of the insert, and an outer trim configured to cover the comfort layer and the insert so that the comfort layer and the insert are positioned between the support layer and the outer trim.

12. The child restraint of claim 11, wherein the insert includes an outer surface configured to engage the head of the child, a rear surface facing toward the support layer, and a chamfered surface arranged at an angle with respect to the rear surface and abutting the comfort layer.

13. The child restraint of claim 11, wherein the first and second side sections spaced apart from a longitudinal centerline of the headrest located between the first and second side sections and extending from the rear section away from the backrest, wherein the outer surface of each insert is spaced a distance from the longitudinal centerline of the headrest, and wherein the distance increases as the outer surface of each insert extends away from the rear section toward a distal end of each corresponding insert.

14. The child restraint of claim 11, wherein each insert is substantially incompressible.

15. The child restraint of claim 14, wherein each insert has a compression strength at 25% of at least 6 psi when measured in accordance with ASTM-D3575.

16. The child restraint of claim 14, wherein the insert includes a foam material.

17. The child restraint of claim 11, wherein each energy-redirection layer has an average density of at least 1 pcf.

18. The child restraint of claim 11, wherein the headrest does not include any side impact air bags.

19. The child restraint of claim 11, wherein the headrest is movable relative to the backrest.

20. The child restraint of claim 11, wherein, at a location along each side section, the outer surface of the comfort layer is arranged at a first angle relative to a plane extending along the rear section and an outer surface of the insert is arranged at a second angle relative to the plane, the second angle being greater than the first angle.